ModelIt!

Student Activity Pack

We Built a Better Dog (And Maybe a Mistake)

How Humans Hack Evolution Through Selective Breeding

Grade 7 | NGSS MS-LS4-5 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **How do you think we got so many different dog breeds from wolves?**

2. **What does it mean to 'breed' plants or animals for specific traits?**

3. **Do you think humans should change organisms to suit our needs? Why or why not?**

4. **What could go wrong if we only breed for one specific trait?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Selection Pressure | Generation Time | Desired Trait Frequency | Genetic Diversity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Mild Selection

Low selection pressure over many generations

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Extreme Purebred

Maximum selection pressure, fast generations

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Wild Population

No selection pressure

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Inbreeding Depression — When closely related organisms breed, harmful recessive genes become more common, causing health problems

GMO Technology — Genetic modification directly edits DNA to insert desired traits, bypassing the slow process of selective breeding

Wild Genetic Reservoir — Wild populations contain genetic diversity that can be crossbred with domesticated populations to restore health

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How have humans changed plants and animals to suit our needs — and at what cost? Use evidence from your simulation to support your answers.

1 How is artificial selection similar to and different from natural selection?

2 Why do many purebred dogs have more health problems than mixed-breed dogs?

3 Is it ethical to breed animals for traits that make them unhealthy but look cute to humans?

4 How has selective breeding of crops like corn and wheat changed human civilization?

5 What are the risks of reducing genetic diversity in our food crops?

8 STEM Challenge: Design a Responsible Breeding Program

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YOUR ENGINEERING MISSION

Design an ethical, evidence-based crop improvement program that increases yield while maintaining genetic diversity, using your understanding of artificial selection.

THE SCENARIO

A farming community is losing crop harvests to a new pest. They need your team to design a selective breeding program that develops pest-resistant crops WITHOUT sacrificing genetic diversity.

Engineering Design Thinking

1. WHAT TRAIT ARE YOU SELECTING FOR AND HOW WILL YOU IDENTIFY IT?
2. HOW WILL YOU MAINTAIN GENETIC DIVERSITY WHILE SELECTING FOR YOUR DESIRED TRAIT?
3. HOW MANY GENERATIONS WILL YOUR PROGRAM NEED TO SEE RESULTS?
4. WHAT ARE THE POTENTIAL RISKS OF YOUR BREEDING PROGRAM?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Geneticists and Agricultural Scientists develop improved crop varieties and animal breeds using selective breeding and biotechnology. They work for universities, USDA, and biotech companies, earning $65,000–$120,000/year.